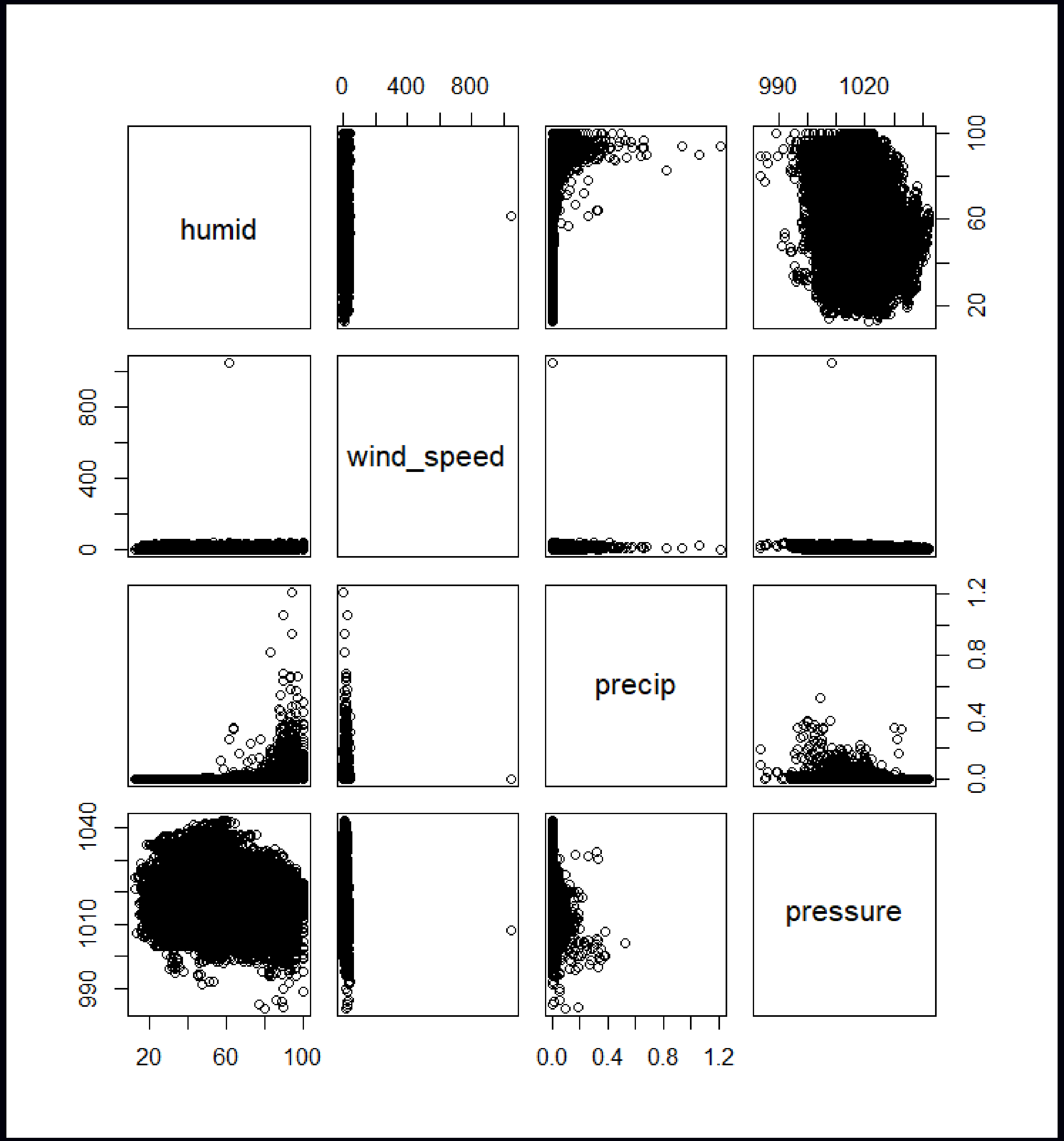
TEMPERATURE PREDICTION USING THE *weather* DATASET

ABOUT THE DATASET

- The dataset contains ourly meterological data for LGA, JFK and EWR.
- There are 15 variables present in this dataset.

OBSERVATIONS FROM THE PAIRWISE SCATTERPLOT



- Humidity and precipitation show a positive relationship.
- Wind speed shows no correlation with any other variables.
- Pressure seems to show a negative correlation with humidity.

MULTIPLE LINEAR REGRESSION

- We are to study the relationship between temperature and other weather variables.
- Predictors are: Humidity,Wind Speed, Precipitation and Pressure
- The fitted model is:

$$\hat{temp} = 697.41 + 0.0229 humid - 0.2399 wind\_speed + 0.3915 precip - 0.6299 pressure$$

- The VIF of the predictors in this model is :

1. Humidity- 1.109859
2. Wind Speed- 1.049394
3. Precipitation- 1.054354
4. Pressure- 1.05047

Hence since all the predictors have VIF less than 5 so multivollinearity is not present.

BACKWARD SUBSET SELECTION

- We start with the full model and remove the least significant predictors
- Model selection based of Adjusted R^2, BIC and Mallow's C_p gives the same model in this case.
- Precipitation is removed from the list of predictors.
- The fitted model is:

$$\hat{temp} = 697.40 + 0.0231\, humid - 0.2397\, windspeed - 0.6299\, pressure$$

- This produces a more simpler and interpretable model.

SHRINKAGE METHODS: LASSO

- LASSO uses the L1 regularization to the regression model.
- It aims at minimising the Penalised Residual Sum of Sqaures(PRSS).
- It performs automatic variable selection by shrinking some coefficients.
- Thus here the fitted model is:

$$\widehat{temp} = 434.47 - 0.0478\, windspeed - 0.3721\, pressure$$

- The optimal λ is selected using k-fold cross validation.
- This λ controls the Bias- Variance tradeoff of the model.
- Here the optimal λ= 1.673418

MODEL COMPARISON

- The fitted models are then used to predict the temperture based on the test data predictor variables.
- The Test MSE for each of the three fitted models is computed.

1. Test MSE(Multiple Linear Regression)=286.583
2. Test MSE(Backward Subset Selection)=286.5805
3. Test MSE(LASSO)=296.3491

CONCLUSION

- LASSO gives a higher Test MSE due to oversimplication.
- LASSO has removed important predictor like humidity which leads to erroneous results.
- Backward Subset Selection has achieved the lowest Test MSE.
- Multiple Linear Regression gives almost identical performance like Backward Subset Selection model.
- Hence, Backward selection emerged as the best approach, providing a simple and interpretable model with optimal predictive performance.
- This analysis highlights that more complex methods do not always lead to better results, especially when the number of predictors is small.